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Question-response representation with dual-level contrastive learning for improving knowledge tracing

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ABSTRACT

Question-response representation (aka. embedding) lies at the core for knowledge tracing, which is pivotal to model students' evolving knowledge states for predicting their future performance. Existing efforts typically learn question-response representations in a supervised manner, thus inevitably neglecting to excavate implicit information existed in student-question-concept connections and suffering from the issue of data sparsity due to rare student-question interactions. Inspired by the recent success of self-supervised learning, in this paper, we propose a **Q**uestion-**R**esponse representation with dual-level **C**ontrastive **L**earning (**QRCL**) for improving knowledge tracing, to address the aforementioned problems. Our approach comprehensively considers three types of views for obtaining question-response representations: the native fold views derived from native interaction between students and questions, the augmented fold views generated by contrastive augmentation with Singular Value Decomposition (SVD), and the co-response relation views acquired by means of correlation among questions. By using delicately designed contrastive rules, Graph Neural Networks (GNNs) are adopted as backbone to encode the information from each view for better generating question-response representation. Extensive experimental results conducted on five datasets clearly show our proposed method has the capability in effectively predicting students' performances.

1. Introduction

Online learning systems, such as Massive Open Online Courses (MOOCs) and Coursera, are developing on an unprecedented scale due to global digitization [1,2]. Diverse personalized intelligent educational services are offered for each student by observing his historical interaction behaviors. As a crucial tool behind the services of these learning platforms, Knowledge Tracing (KT) is defined as the task of automatically discovering the latent knowledge state of different concepts in a dynamic manner. Thereafter, the core of KT centers on predicting the student's performance on future questions according to his answers to a sequence of questions, which can provide customized assistance. More specifically, students can promptly realize their strengths and weaknesses during the learning process. Meanwhile, teachers can design personalized learning programs matching students with different abilities, so as to improve the efficiency of both learning and teaching.

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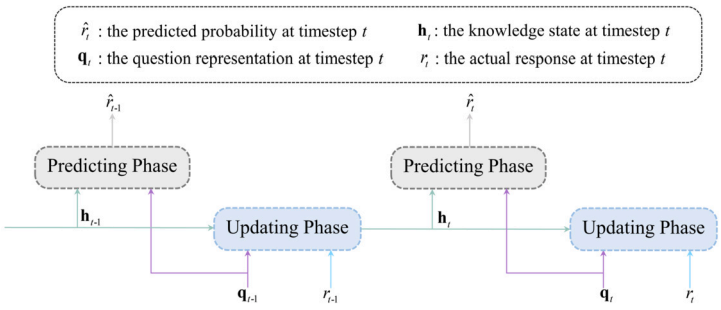


Fig. 1. Two phases of most DNN-based KT methods. The predicting phase aims to predict the student's performance at the current timestep. The updating phase is designed to obtain the knowledge state at the next timestep.

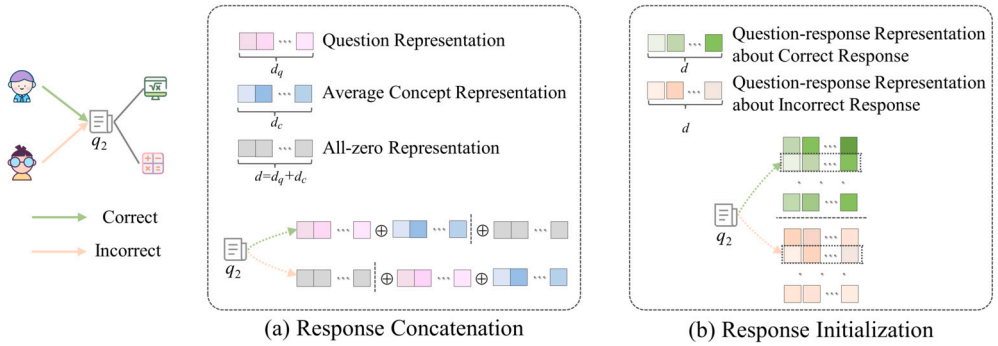


Fig. 2. Two mainstream rules of the construction of question-response representation.

Indeed, there already exist many research efforts relying on Deep Neural Networks (DNNs) to tackle the KT task. Generally speaking, these methods can be divided into two phases as illustrated in Fig. 1: (1) *predicting phase*, which leverages students' current knowledge state and the question representation to generate the predicted probability of answering correctly; (2) *updating phase*, which uses the current knowledge state and question-response representation to acquire the next knowledge state. Hence, high-quality question-response representations play a crucial role in making more accurate prediction. Actually, previous efforts [3–10] have attempted to design rational question-response representation for remarkably promoting the prediction.

In spite of the encouraging progress, the construction of question-response representation depends on one of the following rules (as shown in Fig. 2): (1) response concatenation, such as DKT [3], DHKT [11], CoKT [7], QIKT [12]. As illustrated in Fig. 2(a), the question-response representation of q_2 is encoded as a $2d \times 1$ vector with concatenation of the question representation of q_2 with dimension d_q, the average concept representation related to q_2 with dimension d_c, and a d-dimension all-zero vector, where $d = d_q + d_c$. In particular, concatenation direction (right or left) indicates two responses (correct or incorrect). Slightly different from the above operation is that DKT removes the question representation and its average concept representation is a one-hot vector. (2) response initialization, such as DKVMN [4], SKVMN [13], SAKT [14], CL4KT [8]. As shown in Fig. 2(b), given M questions, the question-response representation of q_2 is generated based on the randomly initialized matrix with dimension $2M \times d$. Although the successful application of these two rules has demonstrated their effectiveness, these question-response representations ignore the underlying information among student-question-concept connections and suffer from sparse supervision signals, owing to the limited student-question interactions. Therefore, the question-response representations learned from the sparse interaction data tend to be readily biased or over-fitted, exerting a negative impact on the accuracy of updating phase to the point of compromising prediction.

Recently, Contrastive learning (CL), as a classical branch of self-supervised learning is one remedy for alleviating the above issue, as its powerful capability in learning discriminative representations by pulling an anchor and a positive sample together while pushing apart the anchor from negative samples [15,16]. In effect, it is non-trivial to introduce contrastive learning into KT task, for that the characteristics of both contrastive learning and knowledge tracing are needed to be carefully considered for balance, which may also bring in the following essential problems: (1) **How to devise proper views for contrastive learning?** Solely augmenting (or corrupting) the native interaction graph as another view is actually distant from sufficient, since it is incapable of adequately utilizing the implicit information (i.e., question-student-question co-response relationship) encapsulated in student-question-concept interaction data. (2) **How to design an appropriate contrastive network to learn high-quality and discriminative question-response representations?** Only performing contrastive learning on the native fold view and augmented fold view cannot reveal sufficient semantic information.

To overcome the above limitations, in this paper, we strive to propose a novel Question-Response representations method with dual-level Contrastive Learning for improving knowledge tracing, named QRCL. Specifically, **to solve the first problem**, we construct three kinds of views, i.e., native fold views, augmented fold views and co-response relation views. Note that the co-response relation

views are generated via exploring the implicit co-response correlation information hidden in student-question-concept connections and serve as complementary views. **To address the second problem**, we further employ Graph Neural Networks (GNNs) as encoders to learn the question-response representations for each question under three types of views. Moreover, at both interaction-level and structure-level contrastive optimization, we subtly design pairs of positive and negative instances to obtain high-quality and discriminative representations and then alleviate the issue of data sparsity. In a nutshell, the main contributions of this work are summarized as follows:

- We emphasize the importance of integrating self-supervised learning into question-response representation in knowledge tracing, which is beneficial for representation learning and further improves the accuracy of final prediction results.
- We propose a dual-level contrastive learning network that obtains the question-response representations at the interaction-level and structure-level respectively, and eventually integrates both to learn question-response representations.
- We conduct extensive experiments on five datasets. The results demonstrate the superiority of QRCL, which shows the effectiveness of dual-level contrastive learning network for improving the prediction performance.

2. Related work

2.1. Knowledge tracing

Knowledge tracing is a popular topic in online learning systems and is traditionally solved using cognitive science and psychology [17,18]. BKT [17], the first applied in the field of education, leverages the Hidden Markov Model which hypothesizes students' knowledge states as a set of binary variables and updates these variables according to Bayesian rules. With the development of Deep Neural Networks, more researchers incorporate deep learning into knowledge tracing and further achieve promising performance. These methods can be largely divided into three types: sequence-aware methods, memory-aware methods and attention-aware methods. To be specific, the sequence-aware methods mainly use well-designed Recurrent Neural Networks (RNNs) to naturally adapt to KT task. For instance, DKT [3] is the first attempt to leverage RNNs to estimate the probability of students correctly answering questions based on their knowledge states. Due to the lack of interpretability of DKT, DKT+ [19] leverages two different regularization terms to augment the original loss function to alleviate above limitation. Though effective, these methods fail to deal with a scenario in which a question may contain multiple concepts. Inspired by memory-augmented neural networks (MANN), the memory-aware approaches are developed to reasonably handle the aforementioned scenario and outperform the sequence-aware methods. DKVMN [4] uses a key-value memory network to store multiple latent concept information and students' knowledge states information. Thereafter, SKVMN [13] improves DKVMN by using a modified Long Short-Term Memory (LSTM) with hops, called Hop-LSTM, to better discover the correlation between latent concepts and questions for prediction. Since these memory-aware methods are unable to capture the dependency relations in interaction data, some attention-based methods are proposed. SAKT [14] designs a self-attention mechanism to capture the relation between students' historical interactions. SAINT [20] utilizes an encoder-decoder model that applies deep self-attentive layers to questions and responses separately to better predict students' performance.

Although these above methods have made a great progress, none of them address the issue of data sparsity and adequately excavate implicit information hidden in student-question-concept connections. Therefore, we strive to develop a reasonable question-response representation method to solve these problems.

2.2. Question representation

Question representation is necessary and essential for predicting performance on future questions during the predicting phase. In general, the question representation can be divided into three categories: index-based methods, text-based methods and relation-based methods. Specifically, the index-based methods explicitly encode the question representation as a one-hot vector by combining index information of questions and students' responses to questions, like DKT [3] and DKT+ [19]. Although these index-based approaches are easy to implement, they cannot consider the textual information in questions. Therefore, some text-based methods are proposed. For instance, RKT [21] and EKT [22] make fully leverage the textual context of questions to empower the question representation. Nevertheless, it is troublesome to extract text information in some scenarios. To address this limitation, several relation-based methods, such as PEBG [23] and SEEP [24], effectively explore the underlying relation information between students-questions-concepts to enhance the question representations. With further research, it is valuable to integrate response information into question representation. Therefore, question-response representation is introduced to assist the knowledge tracing methods to update the student's knowledge state during the updating phase, thus enhancing their prediction performance. Recent question-response representation approaches can be divided into two categories. The first category takes the response concatenation. DKT described in the above index-based methods of question representation is the most fundamental version. Furthermore, DHKT [11] and CoKT [7] obtain the question-response representation with the concatenation of the question representation, the average concept representation that the question covers and the all-zero vector, which is a common way. Moreover, LPKT [25] also takes into account the time information of student's responses in the above concatenation. The second category adopts response initialization, such as DKVMN [4], SAKT [14] and CL4KT [8], they encode the question-response representation according to initialized matrix specifying the embedding dimension.

These two types of methods have proven their effectiveness, but still confront the issue of insufficient interaction information between students and questions. Consequently, we try our best to design a novel question-response representation method for addressing the aforementioned issue.

2.3. Contrastive learning

Contrastive Learning that learns representations by contrastive positive pairs against negative pairs is a special branch of self-supervised learning and yields immense success in diverse domains, including computer vision [26], natural language processing [27] and recommender systems [28]. In graph deep learning, DGI [29], as the first attempt to utilize CL in GNNs, focuses on constructing local node representations and global graph representation as positive pairs to make them the ground truth of each other. Furthermore, GMI [30] is designed to obtain high-quality node representations by contrasting center node against the local ones based on their features and topological structure. More recently, HeCo [31] performs contrastive learning to learn node representations from network schema view and meta-path view.

However, CL in knowledge tracing domain is still in its infancy. Bi-CLKT [32] generates better representations from large amounts of unlabeled data by respectively executing the node-level CL of subgraphs and graph-level CL. CL4KT [8] carefully proposes bidirectional encoders for CL, data augmentation methods and the use of hard negatives to construct meaningful self-supervised signals, so as to improve students' performance prediction. However, CL4KT [8] fails to leverage graph structure information and capture the latent relations among questions. Therefore, S^2-**HHN** [33] develops the intra- and inter-graph mechanisms to aggregate the neighbors' information upon the designed heterogeneous hypergraph network and supplement the supervised prediction task of KT with an auxiliary self-supervised task. Moreover, TCL4KT [34] exhaustively obtains the information on the weighted heterogeneous graph by leveraging three mutually coordinated and supervised view encoders to enhance question representation, thus achieving more accurate predictions. Recently, DTransformer [9] proposes the question-level and knowledge-level framework to explicitly acquire knowledge states and extend their stability with a new training algorithm based on contrastive learning. The aforementioned methods have shown the great potential of contrastive learning on knowledge tracing. Hence, we aim at learning high-quality and discriminative question-response representations with CL to achieve better KT performance.

3. Preliminaries

In this section, we briefly introduce the task of KT with some notations and necessary definitions used in this paper.

Let $S=\{s_1,s_2,\ldots,s_i,\ldots,s_N\}$, $Q=\{q_1,q_2,\ldots,q_j,\ldots,q_M\}$ and $C=\{c_1,c_2,\ldots,c_k,\ldots,c_K\}$, where N is the number of students, M is the number of questions, and K is the number of concepts at a learning system, respectively. Suppose the learning records of all students are $\mathcal{X}=\{\mathcal{X}^{s_1},\mathcal{X}^{s_2},\ldots,\mathcal{X}^{s_N}\}$, where $\mathcal{X}^{s_i}=\{(q_1^{s_i},r_1^{s_i}),(q_2^{s_i},r_2^{s_i}),\ldots,(q_t^{s_i},r_t^{s_i})\}$ are the learning records of student s_i, $q_t^{s_i}$ denotes the question student s_i answers at timestep t and $r_t^{s_i}\in\{1,0\}$ is the corresponding response result. Normally, if he correctly answers question $q_t^{s_i}$, $r_t^{s_i}$ equals to 1, otherwise $r_t^{s_i}$ equals to 0. In particular, $\mathbf{R}_{c_k}\in\{1,0,-1\}^{N\times M}$ indicates the response information of students on questions corresponding to concept c_k, where $\mathbf{R}_{c_k(i,j)}=1/-1$ if student s_i has correct/incorrect response to question q_j, and $\mathbf{R}_{c_k(i,j)}=0$ if student s_i does not give response to question q_j. In addition, $\mathbf{Q}\in\{1,0\}^{M\times K}$ explains the inclusion correlation between questions and concepts labeled by education experts, where $\mathbf{Q}_{(j,k)}=1$ if question q_j includes the concept c_k, otherwise $\mathbf{Q}_{(j,k)}=0$. In most real educational scenarios, a student may answer several questions, and one question contains more than one concept.

For better description, the key notations are summarized in Table 1. In addition, we introduce the following graph information to better illustrate the proposed question-response representation method in our model.

Definition 1 *(Heterogeneous graphs)*. Heterogeneous graphs are denoted as $\mathcal{G}=\left\{\mathcal{G}_{c_k}=<\mathcal{V}_{c_k},\mathcal{E}_{c_k}>\right\}_{k=1}^{k=K}$, where the vertex set $\mathcal{V}_{c_k}\subseteq S\bigcup Q\bigcup\{c_k\}$ and edge set $\mathcal{E}_{c_k}=f\left(S,Q,c_k,\mathbf{R}_{c_k},\mathbf{Q}\right)$. $f(\cdot)$ is the mapping that obtains edges from given vertices and the matrices associated with these vertices. Each heterogeneous graph is established based on the student-question-concept connections under each concept.

Definition 2 *(Native interaction graphs)*. Native interaction graphs related to each concept are derived from the above heterogeneous graphs. To explicitly model the expected question-response representations with respect to correct and incorrect responses, we distinguish two responses of students. Specifically, the native interaction graphs are denoted as $\mathcal{G}^{NI,\mathrm{z}}=\left\{\mathcal{G}_{c_k}^{NI,\mathrm{z}}=<\mathcal{V}_{c_k}^{NI,\mathrm{z}},\mathcal{E}_{c_k}^{NI,\mathrm{z}}>\right\}_{k=1}^{k=K}$, where $\mathcal{V}_{c_k}^{NI,\mathrm{z}}\subseteq S\bigcup Q$, $\mathcal{E}_{c_k}^{NI,\mathrm{z}}=f\left(S,Q,\mathbf{R}_{c_k}^{\mathrm{z}}\right)$, and $\mathrm{z}\in\{+,-\}$ (+/− matches correct/incorrect response respectively). Moreover, the $\mathbf{R}_{c_k}^{+}/\mathbf{R}_{c_k}^{-}$ separately indicates the students select the questions involved the concept c_k and answer them under the correct/incorrect responses. Note that $\mathbf{R}_{c_k}^{\mathrm{z}}$ shares the same dimension as $\mathbf{R}_{c_k}$.

Definition 3 *(Co-response relation graphs)*. To excavate the co-response relationship between arbitrary two questions encapsulated in the learning records, we define $\mathcal{G}^{CR}=\left\{\mathcal{G}_{c_k}^{CR}=<\mathcal{V}_{c_k}^{CR},\mathcal{E}_{c_k}^{CR}>\right\}_{k=1}^{k=K}$ as co-response relation graphs, where $\mathcal{V}_{c_k}^{CR}\subseteq Q$, $\mathcal{E}_{c_k}^{CR}=f\left(Q,\mathbf{Y}_{c_k}^{CR}\right)$. And $\mathbf{Y}_{c_k}^{CR}\in\{1,0,-1\}^{M\times M}$ implies the co-response relationship between two questions. To be specific, $\mathbf{Y}_{c_k(j,m)}^{CR}=1$

Table 1
The key notations and descriptions.

Notation	Description
S	A set of students; $\|S\| = N$; s_i is the ith student
Q	A set of questions; $\|Q\| = M$; q_j is the jth question
C	A set of concepts; $\|C\| = K$; c_k is the kth concept
$\mathcal{X}$	The learning records
$\mathbf{Q}$	The correlation matrix between questions and concepts
$f(\cdot)$	The mapping that obtains edges from given vertices and the matrices associated with these vertices
z	$z = +/-$ matches correct/incorrect responses
$\mathcal{G}^{NF,z}$, $\mathcal{G}^{AF,z}$, $\mathcal{G}^{CR,z}$	The native fold views, augmented fold views and co-response relation views under two responses
GNN_{int}, GNN_{str}	The interaction-level encoder and structure-level encoder
$\mathbf{z}^{Int_NF,z}_{c_k,q_j}$, $\mathbf{z}^{AF,z}_{c_k,q_j}$	The latent question-response representations of q_j in native fold views at interaction-level contrastive learning and augmented fold views under both two responses and concept c_k
$\mathbf{z}^{Str_NF,z}_{c_k,q_j}$, $\mathbf{z}^{CR,z}_{c_k,q_j}$	The latent question-response representations of q_j in native fold views at structure-level contrastive learning and co-response relation views under both two responses and concept c_k
ℓ^{Int}, ℓ^{Str}	The interaction-level loss and structure-level loss
$\bar{\mathbf{z}}^{z}_{q_j}$	The ultimate learned question-response representations of q_j under two responses

if the weight $w^{+}_{c_k\,q_j,q_m} / w^{+}_{c_k\,q_m,q_j} > \eta$; $\mathbf{Y}^{CR}_{c_k(j,m)} = -1$ if the weight $w^{-}_{c_k\,q_j,q_m} / w^{-}_{c_k\,q_m,q_j} > \eta$ otherwise $\mathbf{Y}^{CR}_{c_k(j,m)} = 0$. Here, η is a pre-defined threshold. Therefore, the calculation of $w^{z}_{c_k\,q_j,q_m}$ under two responses can be formulated as follows:

$$w^{z}_{c_k\,q_j,q_m} = \frac{F^{z,r}_{c_k}\left(q_j, q_m\right)}{\sum_{j'=1}^{M} F^{z,o}_{c_k}\left(q_j, q_{j'}\right) + 0.01} \tag{1}$$

where $w^{z}_{c_k\,q_j,q_m}$ measures the co-response rate of question q_j and q_m related to both concept c_k and two responses. $F^{z,r}_{c_k}$ counts the number of co-response of two questions that are answered by the same student with a consistent response (i.e., consistently correct or incorrect response). $F^{z,o}_{c_k}$ counts the number of co-occurrence of two questions involving the question q_j. Furthermore, we introduce the smoothing parameter 0.01 to prevent the denominator from being too small [35].

4. Method

In this section, we present the proposed QRCL, which aims to exploit the potential information existed in student-question-concept connections and deal with the issue of data sparsity in a self-supervised learning scheme as illustrated in Fig. 3. And the details of QRCL are introduced as follows.

4.1. View generation

It is of great necessity to select suitable and proper views for empowering the contrastive learning later on [36]. Since generating a contrastive view using only the augmentation method is not adequate, we also consider the significant and underlying information, i.e., co-response relation between questions. As Fig. 3(b) displays, the designed model comprises three kinds of views related to a particular concept, including the native fold views, augmented fold views and co-response relation views under two responses. The specific methods for each view generation are detailed below.

4.1.1. Native and augmented fold views

In this subsection, we first introduce the native fold views. Based on the definition of native interaction graphs, we denote the native fold views as $\mathcal{G}^{NF,z} = \left\{ \mathcal{G}^{NF,z}_{c_k} = < \mathcal{V}^{NF,z}_{c_k}, \mathcal{E}^{NF,z}_{c_k} > \right\}_{k=1}^{k=K}$ to characterize the common interaction of two questions with different concepts under two responses, where $\mathcal{V}^{NF,z}_{c_k} \subseteq Q$, $\mathcal{E}^{NF,z}_{c_k} = f\left(Q, \mathbf{Y}^{NF,z}_{c_k}\right)$ and $\mathbf{Y}^{NF,z}_{c_k} \in \mathbb{R}^{M \times M}$. In particular, $\mathbf{Y}^{NF,z}_{c_k\,(j,m)}$ reveals that both question q_j and q_m are interacted by the same students in $\mathcal{G}^{NF,z}_{c_k}$. As Fig. 4 illustrates, we separately built edges between any two questions (i.e., edge between q_1 and q_3, edge between q_1 and q_4, and edge between q_3 and q_4) since student s_1 correctly answered questions q_1, q_3 and q_4. Moreover, the edge between q_2 and q_3 is constructed based on the responses of student s_2. It is worth mentioning that this view will be used both in subsequent interaction-level and structure-level contrastive learning.

Next, we follow the same rule of LightGCL [37] to reconstruct the native interaction graphs with SVD scheme instead of random perturbation to obtain augmented interaction graphs, avoiding misleading signals with random augmenting perturbation graphs. The key reason for choosing the SVD scheme is that the augmented interaction graphs are created with preserving global student-question

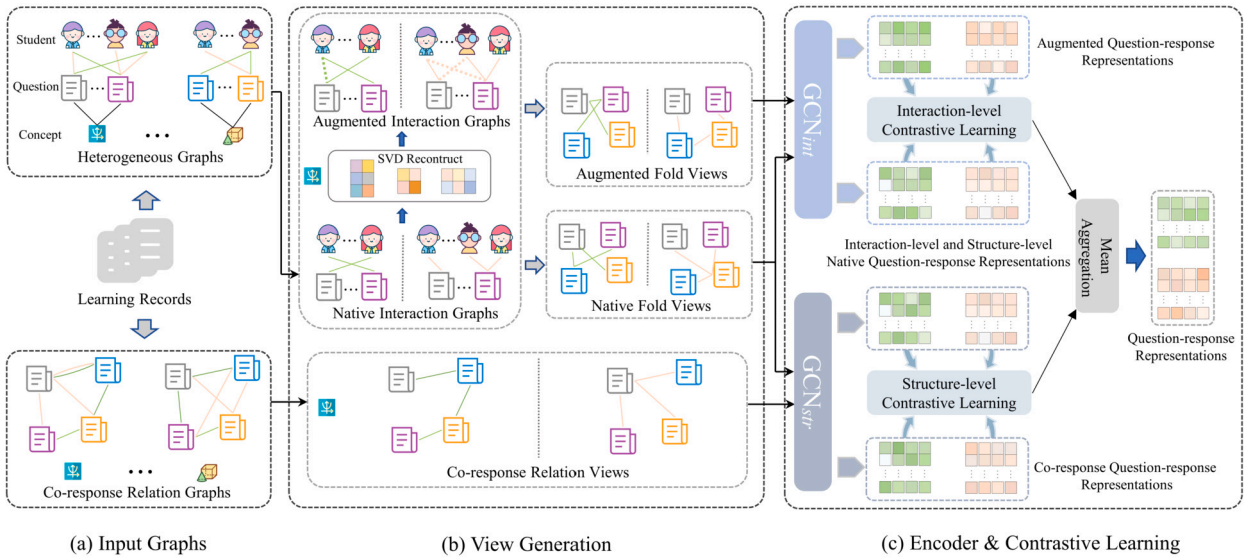


Fig. 3. The framework of QRCL. (a) Input graphs module introduces the heterogeneous graphs and co-response relation graphs obtained from the learning records. Note that, for heterogeneous graphs, the black line indicates the questions involved by a certain concept. The green and orange lines imply the students' correct and incorrect responses to the questions respectively. For co-response relation graphs, the green and orange lines depict the co-response relations between the questions in the case of correct and incorrect responses, respectively. (b) View generation module first shows the native interaction graphs, and further constructs the augmented interaction graphs with SVD scheme. It is worth noting that the green and orange dotted lines in the augmented interaction graphs indicate the newly generated edges under correct and incorrect responses, respectively. (c) Encoder and contrastive learning module contains two stages. The first stage is to derive the corresponding initial question-response embeddings from the GNN_{int} and GNN_{str} encoders. The second stage is to perform dual-level contrastive learning on these embeddings and finally attain the ultimate question-response representation.

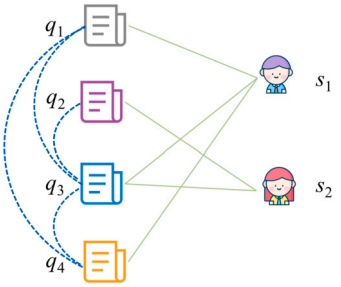


Fig. 4. An example of constructing fold view. Student s_1 correctly answers questions q_1, q_3 and q_4. Student s_2 correctly answers questions q_2, q_3. The green solid lines indicate the students' correct responses to the questions. The blue dotted lines imply that two questions were answered by the same student.

interaction information, which effectively mitigates the issue of student-question interaction data sparsity and further enhances the ultimate question-response representations.

Specifically, we perform SVD on $\mathbf{R}^{\mathrm{z}}_{c_k}$ of native interaction graph with the following form:

$$\mathbf{N}^{\mathrm{z}}_{c_k}, \mathbf{\Sigma}^{\mathrm{z}}_{c_k}, \mathbf{K}^{\mathrm{z}}_{c_k} = \mathrm{SVD}\left(\mathbf{R}^{\mathrm{z}}_{c_k}, \upsilon\right) \tag{2}$$

where $\mathbf{N}^{\mathrm{z}}_{c_k} \in \mathbb{R}^{N\times\upsilon}$ is the decompose student matrix, $\mathbf{\Sigma}^{\mathrm{z}}_{c_k} \in \mathbb{R}^{\upsilon\times\upsilon}$ is the diagonal matrix of singular values, $\mathbf{K}^{\mathrm{z}}_{c_k} \in \mathbb{R}^{\upsilon\times M}$ is the decomposed question matrix, and υ is the number of largest singular values associated with the principal components of the matrix. After that, we attain the reconstructed matrix $\mathbf{R}^{AI,\mathrm{z}}_{c_k} \in \mathbb{R}^{N\times M}$ in the augmented interaction graphs as follows:

$$\mathbf{R}^{AI,\mathrm{z}}_{c_k} = \mathbf{N}^{\mathrm{z}}_{c_k} \mathbf{\Sigma}^{\mathrm{z}}_{c_k} \mathbf{K}^{\mathrm{z}}_{c_k} \tag{3}$$

Thus, we acquire the augmented interaction graphs $\mathcal{G}^{AI,\mathrm{z}} = \left\{\mathcal{G}^{AI,\mathrm{z}}_{c_k} =< \mathcal{V}^{AI,\mathrm{z}}_{c_k}, \mathcal{E}^{AI,\mathrm{z}}_{c_k} >\right\}^{k=K}_{k=1}$, where $\mathcal{V}^{AI,\mathrm{z}}_{c_k} \subseteq S \bigcup Q$ and $\mathcal{E}^{AI,\mathrm{z}}_{c_k} = f\left(S, Q, \mathbf{R}^{AI,\mathrm{z}}_{c_k}\right)$. Similar to the generation process of native fold views, we can get the augmented fold views as $\mathcal{G}^{AF,\mathrm{z}} = \left\{\mathcal{G}^{AF,\mathrm{z}}_{c_k} =< \mathcal{V}^{AF,\mathrm{z}}_{c_k}, \mathcal{E}^{AF,\mathrm{z}}_{c_k} >\right\}^{k=K}_{k=1}$, where $\mathcal{V}^{AF,\mathrm{z}}_{c_k} \subseteq Q$ and $\mathcal{E}^{AF,\mathrm{z}}_{c_k} = f\left(Q, \mathbf{Y}^{AF,\mathrm{z}}_{c_k}\right)$.

4.1.2. Co-response relation views

Adequate learning records of students are useful information but have not been sufficiently explored. Fortunately, we indirectly consider the correct and incorrect responses hidden in the students' learning records from the perspective of co-response between different questions, which explores underlying information existed in student-question-concept associations to a certain extent. Specifically, we partition two distinct responses in co-response relation graphs $\mathcal{G}^{CR}$ and natively obtain the co-response relation views $\mathcal{G}^{CR,z} = \left\{ \mathcal{G}_{c_k}^{CR,z} =< \mathcal{V}_{c_k}^{CR,z}, \mathcal{E}_{c_k}^{CR,z} > \right\}_{k=1}^{k=K}$, where $\mathcal{V}_{c_k}^{CR,z} \subseteq Q$ and $\mathcal{E}_{c_k}^{CR,z} = f\left(Q, \mathbf{Y}_{c_k}^{CR,z}\right)$.

Up to now, the native fold views $\mathcal{G}^{NF,z}$, augmented fold views $\mathcal{G}^{AF,z}$, and co-response relation views $\mathcal{G}^{CR,z}$ under all concepts are successfully generated. The following interaction-level and structure-level contrastive learning are performed on the above three types of views, which will be illustrated in detail.

4.2. Interaction-level contrastive network

With the help of the above obtained native and augmented fold views, we move on to explore two graph views with proper graph encoder and perform contrastive learning between them to supervise each other.

4.2.1. Interaction-level view encoder

Our goal is to acquire question-response representations using an interaction-level contrastive learning approach that considers the high-order connections between different questions and their respective responses. This approach will be applied to both the native and augmented fold views to better capture the dependencies between questions. Here, we leverage LightGCN [38] to encode collaborative signals from these two views owing to its simple message propagation and aggregation mechanism. Note that, we name the encoder here as GNN_{int} to update both the interaction-level native question-response representations and augmented question-response representations.

Specifically, we first randomly initialize the question-response representations by the defined two trainable matrices $\mathbf{E}_{c_k}^{Int_NF,z} \in \mathbb{R}^{M\times d}$ related to concept c_k for native fold views, where d is the embedding dimension of questions. To be specific, $\mathbf{E}_{c_k}^{Int_NF,+}/\mathbf{E}_{c_k}^{Int_NF,-}$ separately indicates the initialized question-response representations matching correct/incorrect responses. Note that for brevity we replace $\mathbf{E}_{c_k}^{Int_NF,z}$ with $\mathbf{E}^{Int_NF}$ to specify the scenario under both concept c_k and two responses in subsequent statements. Consequently, the initial embeddings $\mathbf{e}_{q_j}^{Int_NF} \in \mathbb{R}^d$ corresponding to the particular question q_j are obtained as follows:

$$\mathbf{e}_{q_j}^{Int_NF} = \left(\mathbf{E}^{Int_NF}\right)^T \tilde{\mathbf{e}}_{q_j}^{Int_NF} \tag{4}$$

where $\tilde{\mathbf{e}}_{q_j}^{Int_NF} \in \mathbb{R}^M$ is the one-hot embedding of q_j. In the $(l+1)$-th layers, the aggregation proceeding is formulated as below:

$$\mathbf{e}_{q_j}^{Int_NF,(l+1)} = \sum_{q_m \in \mathcal{N}_{q_j}^{Int_NF}} \frac{1}{\sqrt{\left|\mathcal{N}_{q_j}^{Int_NF}\right|\left|\mathcal{N}_{q_m}^{Int_NF}\right|}} \mathbf{e}_{q_m}^{Int_NF,(l)} \tag{5}$$

where $\mathcal{N}_{q_j}^{Int_NF}$ and $\mathcal{N}_{q_m}^{Int_NF}$ are the set of connected neighbors of q_j and q_m under native fold views, two responses and the specific concept c_k respectively. Note that $\mathbf{e}_{q_j}^{Int_NF,(0)} = \mathbf{e}_{q_j}^{Int_NF}$. Afterwards, we further combine all layers' representations through the message propagation of L layers as follows:

$$\bar{\mathbf{e}}_{q_j}^{Int_NF} = \sum_{l=0}^{L} \alpha_l \mathbf{e}_{q_j}^{Int_NF,(l)} \tag{6}$$

where $\alpha_l > 0$ denotes the contribution of the l-th layer in constructing final representation. It is a hyperparameter and can be tuned manually. Following the LightGCN, we empirically find that setting α_l as $1/(L+1)$ achieves outstanding performance in general.

Analogously, we first denote the question-response representations for augmented fold views by two initialized trainable matrices $\mathbf{E}^{AF} \in \mathbb{R}^{M\times d}$. Next, the initial embeddings $\mathbf{e}_{q_j}^{AF}$ of question q_j corresponding to two responses can be obtained by borrowing from Eq. (4). Lastly, we use GNN_{int} formulated above to obtain the ultimate embedding $\bar{\mathbf{e}}_{q_j}^{AF}$ of all layers' aggregation of q_j.

4.2.2. Interaction-level contrastive optimization

With the above view-specific embeddings $\bar{\mathbf{e}}_{q_j}^{Int_NF}$ and $\bar{\mathbf{e}}_{q_j}^{AF}$ for question q_j from native and augmented fold views, interaction-level contrastive learning is executed for supervising two views to acquire discriminative question-response representations. Furthermore, we map these representations to a uniform latent space via MultiLayer Perceptron (MLP) and they can be computed as follows:

$$\mathbf{z}_{q_j}^{Int_NF} = \text{MLP}\left(\bar{\mathbf{e}}_{q_j}^{Int_NF}\right) \tag{7}$$

$$\mathbf{z}_{q_j}^{AF} = \text{MLP}\left(\bar{\mathbf{e}}_{q_j}^{AF}\right) \tag{8}$$

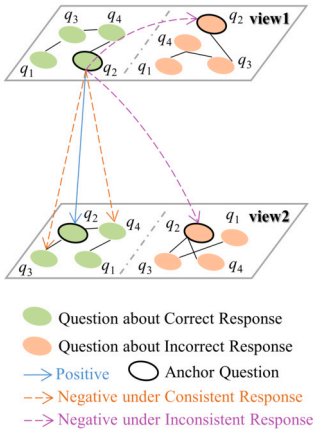


Fig. 5. An Example of Contrastive Learning.

So far, we obtain the latent question-response representations $\mathbf{z}_{q_j}^{Int_NF}$ and $\mathbf{z}_{q_j}^{AF}$ from the native fold views and augmented fold views, respectively. To clearly formulate the loss of consequent interaction-level contrastive learning, we restore the original notation and thus replace $\mathbf{z}_{q_j}^{Int_NF}$ and $\mathbf{z}_{q_j}^{AF}$ with $\mathbf{z}_{c_k,q_j}^{Int_NF,z}$ and $\mathbf{z}_{c_k,q_j}^{AF,z}$. After that, we introduce the crucial positive and negative sampling strategy. As Fig. 5 illustrates, given an anchor question q_2, question-response representation of this question learned by another contrastive view under the consistent response forms the positive samples (i.e., question q_2 pointed by a blue solid line). It is worth mentioning that negative samples are composed of two sources (i.e., consistent response and inconsistent response). For negative samples under consistent response, we select the question-response representations of first-order neighbors of this question as them (i.e., question q_3 and q_4 pointed by two orange dashed lines). The negative samples under inconsistent response are the same question-response representations from two views (i.e., two questions q_2 pointed by two purple dashed lines). The intuition behind the above strategy of negative samples selection under inconsistent response is that although it is the same question about two responses, the learned question-response representations with respect to two responses should be slightly different.

Normally, we leverage InfoNCE [39] to learn informative and discriminative question-response representations by pulling similar positive samples together and pushing negative samples apart. Therefore, given a question q_j related to correct response, we have the following contrastive loss:

$$\ell_{c_k,q_j}^{Int_NF,+} = -\log \frac{\phi\left(\mathbf{z}_{c_k,q_j}^{Int_NF,+}, \mathbf{z}_{c_k,q_j}^{AF,+}\right)}{\underbrace{\phi\left(\mathbf{z}_{c_k,q_j}^{Int_NF,+}, \mathbf{z}_{c_k,q_j}^{Int_NF,-}\right) + \phi\left(\mathbf{z}_{c_k,q_j}^{Int_NF,+}, \mathbf{z}_{c_k,q_j}^{AF,-}\right)}_{\text{Inconsistent response}} + \underbrace{\sum_{q_m \in \mathcal{N}_{c_k,q_j}^{AF,+}} \phi\left(\mathbf{z}_{c_k,q_j}^{Int_NF,+}, \mathbf{z}_{c_k,q_m}^{AF,+}\right)}_{\text{Consistent response}}} \tag{9}$$

$$\ell_{c_k,q_j}^{AF,+} = -\log \frac{\phi\left(\mathbf{z}_{c_k,q_j}^{AF,+}, \mathbf{z}_{c_k,q_j}^{Int_NF,+}\right)}{\underbrace{\phi\left(\mathbf{z}_{c_k,q_j}^{AF,+}, \mathbf{z}_{c_k,q_j}^{AF,-}\right) + \phi\left(\mathbf{z}_{c_k,q_j}^{AF,+}, \mathbf{z}_{c_k,q_j}^{Int_NF,-}\right)}_{\text{Inconsistent response}} + \underbrace{\sum_{q_m \in \mathcal{N}_{c_k,q_j}^{Int_NF,+}} \phi\left(\mathbf{z}_{c_k,q_j}^{AF,+}, \mathbf{z}_{c_k,q_m}^{Int_NF,+}\right)}_{\text{Consistent response}}} \tag{10}$$

where $\phi\left(\mathbf{x}_1, \mathbf{x}_2\right) = \exp\left[sim\left(\mathbf{x}_1, \mathbf{x}_2\right)/\tau\right]$, where $sim(\cdot)$ denotes the cosine similarity calculating between two vectors and τ is a temperature hyperparameter. In addition, $\mathcal{N}_{c_k,q_j}^{AF,+}$ denotes the set of first-order neighbors of question q_j in augmented fold views related to correct response and concept c_k. $\mathcal{N}_{c_k,q_j}^{Int_NF,+}$ means the set of first-order neighbors of question q_j in native fold views at interaction-level contrastive learning related to correct response and concept c_k. And the contrastive loss $\ell_{c_k,q_j}^{Int_NF,-}/\ell_{c_k,q_j}^{AF,-}$ calculating from incorrect response is similar as $\ell_{c_k,q_j}^{Int_NF,+}/\ell_{c_k,q_j}^{AF,+}$. Then, the interaction-level contrastive loss can be formulated as follows:

$$\ell^{Int} = \frac{1}{4K}\sum_{k=1}^{K}\sum_{j=1}^{M}\sum_{z\in\{+,-\}}\left(\ell_{c_k,q_j}^{Int_NF,z} + \ell_{c_k,q_j}^{AF,z}\right) \tag{11}$$

It is worth noting that when fixing a certain concept and question, there will be correct and incorrect question-response representations under each view.

4.3. Structure-level contrastive network

Although the interaction information between students and questions has been revealed from interaction-level views (i.e., native fold views at interaction-level contrastive learning and augment fold views), the co-response semantic information between questions hasn't been explored. Therefore, we introduce the structure-level contrastive network to excavate the implicit co-response semantic information to complement the single interaction information between students and questions, which eventually enhances the ultimate question-response representations.

4.3.1. Structure-level view encoder

In this section, we intend to generate the question-response representations from native fold views at structure-level contrastive learning and co-response relation views, different concepts and two responses. Here, we choose the similar architecture to interaction-level view encoder, named GNN_{str}, for updating the structure-level native question-response representations and co-response question-response representations. Equally, for a question q_j, two initial embeddings of q_j are denoted as $\mathbf{e}_{q_j}^{Str_NF}$ with two initialized trainable matrices $\mathbf{E}^{Str_NF} \in \mathbb{R}^{M\times d}$ and Eq. (4). Moreover, we formulated a different aggregation process in the $(l+1)$-th layers as follows:

$$\mathbf{e}_{q_j}^{Str_NF,(l+1)} = \mathbf{e}_{q_j}^{Str_NF,(l)} + \sum_{q_m \in \mathcal{N}_{q_j}^{Str_NF}} \frac{1}{\sqrt{\left|\mathcal{N}_{q_j}^{Str_NF}\right|\left|\mathcal{N}_{q_m}^{Str_NF}\right|}} \mathbf{e}_{q_m}^{Str_NF,(l)} \tag{12}$$

where $\mathbf{e}_{q_j}^{Str_NF,(0)} = \mathbf{e}_{q_j}^{Str_NF}$. Next, we can aggregate all layers' representations through the message propagation mechanism in Eq. (6) as $\bar{\mathbf{e}}_{q_j}^{Str_NF}$. Similarly, we first denote the question-response representations for co-response relation views by the two initialized trainable matrices $\mathbf{E}^{CR} \in \mathbb{R}^{M\times d}$. Next, the initial embeddings $\mathbf{e}_{q_j}^{CR}$ of question q_j corresponding to two different responses can be obtained via Eq. (4). Finally, GNN_{str} formulated above is utilized to obtain the ultimate embeddings $\bar{\mathbf{e}}_{q_j}^{CR}$ of all layers' aggregation of q_j.

4.3.2. Structure-level contrastive optimization

After obtaining the question-response representations from native fold views at structure-level contrastive learning and co-response relation views, we can denote them as $\mathbf{z}_{c_k,q_j}^{Str_NF,\mathrm{z}}$ and $\mathbf{z}_{c_k,q_j}^{CR,\mathrm{z}}$ by mapping them into a uniform latent space on the basis of Eq. (7) and Eq. (8). Besides, we recover the original symbol, thus replacing $\mathbf{z}_{q_j}^{Str_NF}$ and $\mathbf{z}_{q_j}^{CR}$ with $\mathbf{z}_{c_k,q_j}^{Str_NF,\mathrm{z}}$ and $\mathbf{z}_{c_k,q_j}^{CR,\mathrm{z}}$. Subsequently, with the same positive and negative sampling strategy to interaction-level contrastive optimization, we define the following contrastive loss:

$$\ell_{c_k,q_j}^{Str_NF,+} = -\log \frac{\phi\left(\mathbf{z}_{c_k,q_j}^{Str_NF,+}, \mathbf{z}_{c_k,q_j}^{CR,+}\right)}{\underbrace{\phi\left(\mathbf{z}_{c_k,q_j}^{Str_NF,+}, \mathbf{z}_{c_k,q_j}^{Str_NF,-}\right) + \phi\left(\mathbf{z}_{c_k,q_j}^{Str_NF,+}, \mathbf{z}_{c_k,q_j}^{CR,-}\right)}_{\text{Inconsistent response}} + \underbrace{\sum_{q_m \in \mathcal{N}_{c_k,q_j}^{CR,+}} \phi\left(\mathbf{z}_{c_k,q_j}^{Str_NF,+}, \mathbf{z}_{c_k,q_m}^{CR,+}\right)}_{\text{Consistent response}}} \tag{13}$$

$$\ell_{c_k,q_j}^{CR,+} = -\log \frac{\phi\left(\mathbf{z}_{c_k,q_j}^{CR,+}, \mathbf{z}_{c_k,q_j}^{Str_NF,+}\right)}{\underbrace{\phi\left(\mathbf{z}_{c_k,q_j}^{CR,+}, \mathbf{z}_{c_k,q_j}^{CR,-}\right) + \phi\left(\mathbf{z}_{c_k,q_j}^{CR,+}, \mathbf{z}_{c_k,q_j}^{Str_NF,-}\right)}_{\text{Inconsistent response}} + \underbrace{\sum_{q_m \in \mathcal{N}_{c_k,q_j}^{Str_NF,+}} \phi\left(\mathbf{z}_{c_k,q_j}^{CR,+}, \mathbf{z}_{c_k,q_m}^{Str_NF,+}\right)}_{\text{Consistent response}}} \tag{14}$$

where $\mathcal{N}_{c_k,q_j}^{CR,+}$ is the collection of first-order neighbors of question q_j in co-response relation views related to correct response and concept c_k. $\mathcal{N}_{c_k,q_j}^{Str_NF,+}$ indicates the collection of first-order neighbors of question q_j in native fold views at structure-level contrastive learning related to correct response and concept c_k. Note that we can also calculate the contrastive loss $\ell_{c_k,q_j}^{Str_NF,-}/\ell_{c_k,q_j}^{CR,-}$ from incorrect response by using the same way as described in $\ell_{c_k,q_j}^{Str_NF,+}/\ell_{c_k,q_j}^{CR,+}$. Next, we have the structure-level contrastive loss formulated as follows:

$$\ell^{Str} = \frac{1}{4K}\sum_{k=1}^{K}\sum_{j=1}^{M}\sum_{\mathrm{z}\in\{+,-\}}\left(\ell_{c_k,q_j}^{Str_NF,\mathrm{z}} + \ell_{c_k,q_j}^{CR,\mathrm{z}}\right) \tag{15}$$

4.4. Joint contrastive loss optimization

To better learn the question-response representations, we combine the interaction-level contrastive loss with structure-level contrastive loss and further define the overall contrastive loss objective as follows:

$$\ell = \alpha \ell^{Int} + (1-\alpha)\ell^{Str} + \lambda \|\theta\|_2^2 \tag{16}$$

where α is the coefficient adjusting the balance between these two losses, θ is the set of model parameter, λ controls L_2 regularization term. By minimizing the above objective function, we can obtain the multiple effective question-response representations about all concepts and two responses.

4.5. Question-response representation acquisition

With the dual-level contrastive optimization, we derive multiple trained question-response representations for a particular question q_j from interaction-level and structure-level perspectives. Specifically, from interaction-level perspective, the question-response representations of q_j are denoted $\mathbf{z}_{c_k,q_j}^{Int_NF,z}$ and $\mathbf{z}_{c_k,q_j}^{AF,z}$. From structure-level perspective, its question-response representations are $\mathbf{z}_{c_k,q_j}^{Str_NF,z}$ and $\mathbf{z}_{c_k,q_j}^{CR,z}$, respectively.

After that, we need to aggregate the question-response representations from dual-level contrastive learning. Since each question covers different concepts, we look forward to combining the corresponding question-response representations under the concepts related to the question. Taking $\mathbf{z}_{c_k,q_j}^{Int_NF,z}$ as an example, we have the following aggregation process:

$$\mathbf{z}_{q_j}^{Int_NF,z} = \frac{1}{a}\sum_{k=1}^{K}\left[\mathbb{I}\left(\mathbf{Q}_{(j,k)}=1\right)\times \mathbf{z}_{c_k,q_j}^{Int_NF,z}\right] \tag{17}$$

where $a = \sum_{k=1}^{K}\mathbb{I}\left(\mathbf{Q}_{(j,k)}=1\right)$, counting the number of concepts included in question q_j. $\mathbb{I}(\cdot)$ is an indicator function that equals to 1 if condition is satisfied otherwise equals to 0. Analogously, we can obtain $\mathbf{z}_{q_j}^{AF,z}$, $\mathbf{z}_{q_j}^{Str_NF,z}$ and $\mathbf{z}_{q_j}^{CR,z}$. Here, we omit the details for brevity.

Eventually, considering the question-response representations learned from three kinds of views have the same significance, we naturally perform the mean aggregation to generate the ultimate comprehensive question-response representations of q_j as:

$$\bar{\mathbf{z}}_{q_j}^{z} = \mathrm{MEAN}\left(\mathbf{z}_{q_j}^{Int_NF,z}, \mathbf{z}_{q_j}^{AF,z}, \mathbf{z}_{q_j}^{Str_NF,z}, \mathbf{z}_{q_j}^{CR,z}\right) \tag{18}$$

where $\mathrm{MEAN}(\cdot)$ is the operation of mean pooling. It is worth noting that $\bar{\mathbf{z}}_{q_j}^{z}$ can be utilized to better update the knowledge state of student, which is an essential and crucial component for KT models, so as to improve the performance of a large set of downstream KT models.

Algorithm 1 sums up our QRCL method.

Algorithm 1 The QRCL method.

Input: The set of students S, the set of questions Q, the set of concepts C, the learning records $\mathcal{X}$, and the correlation matrix between questions and concepts $\mathbf{Q}$

Output: The question-response representations $\bar{\mathbf{z}}_{q_j}^{z}$.

Construct the heterogeneous graphs $\mathcal{G}$ and co-response relation graphs $\mathcal{G}^{CR}$ under different concepts according to $\mathcal{X}$ and $\mathbf{Q}$;
Obtain the native interaction graphs $\mathcal{G}^{NI,z}$ under different concepts and two responses via $\mathcal{G}$;
for $k=1$ to K
  Given the $\mathcal{G}_{c_k}^{NI,z}$ in $\mathcal{G}^{NI,z}$, attain the two augmented interaction graphs $\mathcal{G}_{c_k}^{AI,z}$ under concept c_k and two responses according to Eq. (2) and Eq. (3);
  Acquire the two native fold views $\mathcal{G}_{c_k}^{NF,z}$ and augmented fold views $\mathcal{G}_{c_k}^{AF,z}$ under concept c_k and two responses through $\mathcal{G}_{c_k}^{NI,z}$ and $\mathcal{G}_{c_k}^{AI,z}$ respectively;
  Obtain the two co-response relation views $\mathcal{G}_{c_k}^{CR,z}$ by means of $\mathcal{G}_{c_k}^{CR}$ in $\mathcal{G}^{CR}$;
  for $j=1$ to M
    Based on GNN_{int}, attain the initialized question-response representations $\bar{\mathbf{e}}_{q_j}^{Int_NF}$ and $\bar{\mathbf{e}}_{q_j}^{AF}$ of question q_j in native fold views at interaction-level contrastive learning and augmented fold views;
    Based on GNN_{str}, attain the initialized question-response representations $\bar{\mathbf{e}}_{q_j}^{Str_NF}$ and $\bar{\mathbf{e}}_{q_j}^{CR}$ of question q_j in native fold views at structure-level contrastive learning and co-response relation views;
    Obtain the latent interaction-level native question-response representations $\mathbf{z}_{c_k,q_j}^{Int_NF,z}$ and augmented question-response representations $\mathbf{z}_{c_k,q_j}^{AF,z}$ of question q_j in step 8 according to Eq. (7) and Eq. (8). Analogously, obtain the latent structure-level native question-response representations $\mathbf{z}_{c_k,q_j}^{Str_NF,z}$ and co-response question-response representations $\mathbf{z}_{c_k,q_j}^{CR,z}$ of question q_j in step 9;
    Calculate the loss $\ell_{c_k,q_j}^{Int_NF,z}/\ell_{c_k,q_j}^{AF,z}$ of question q_j under concept c_k and two responses at interaction-level contrastive learning by Eq. (9) and Eq. (10). Moreover, attain the loss $\ell_{c_k,q_j}^{Str_NF,z}/\ell_{c_k,q_j}^{CR,z}$ of question q_j under concept c_k and two responses at structure-level contrastive learning by Eq. (13) and Eq. (14);
  end for;
end for;
Calculate the interaction-level contrastive loss ℓ^{Int} and structure-level contrastive loss ℓ^{Str} according to Eq. (11) and Eq. (15);
Update the parameters of model with backpropagation algorithm until convergence by Eq. (16);
Obtain the final question-response representations $\bar{\mathbf{z}}_{q_j}^{z}$ of question q_j under two responses according to Eq. (17) and Eq. (18).

4.6. Time complexity analysis

QRCL generates question-response representations through dual-level contrastive learning among three types of views, employing carefully designed contrastive rules. To precisely understand this trade-off, we conduct a time complexity analysis at a constant level. In this analysis, we exclude the time complexity of view generation as it serves as a preprocessing step for subsequent model learning. Consequently, our focus lies solely on the encoder and contrastive learning module, including the interaction-level contrastive network, structure-level contrastive network, and question-response representation acquisition. This allows for a more accurate evaluation of the overall time complexity. For the interaction-level contrastive network, the message propagation and aggregation (Eq. (4)-(6)) are the crucial operations, costing $O(Nd + ldM + l)$, where N and M denote the number of students and questions respectively. d is the embedding size and l is the number of layers of convolution. Besides, suppose the number of neighbors of the anchor question is a, and the time complexity of interaction-level contrastive loss (Eq. (9)-(11)) is $O(4aKM)$, where K is the number of questions. Analogously, for the structure-level contrastive network (Eq. (12)-(15)), the total complexity is $O(Nd + ldM + l + 4aKM)$. For the question-response representation acquisition (Eq. (17)-(18)), $O(M(K + d))$ is the time complexity of generating the ultimate comprehensive question-response representations of all questions. Eventually, the total time complexity of QRCL is $O(2Nd + 2ldM + 2l + 8aKM + MK + Md)$.

5. Experiments

In this section, we conduct extensive experiments to estimate the performance of our proposed QRCL method. Specifically, we first introduce the experimental setup. After that, we demonstrate the effectiveness and interpretability of our method by answering the following research questions:

- **RQ1**: How do the response representations generated by QRCL improve the performance of baselines?
- **RQ2**: What is the influence of two crucial components (i.e., augmented fold views and co-response relation views) of QRCL?
- **RQ3**: How does the model perform in terms of efficiency?
- **RQ4**: How do the significant hyperparameters affect QRCL effectiveness?
- **RQ5**: How about the interpretation of question-response representations obtained by QRCL method?

5.1. Experiment setup

5.1.1. Datasets

We verify the effectiveness of our method on the following five real-world datasets, i.e., ASSIST09,[1] ASSIST12,[2] ASSISTchall,[3] Statics2011[4] and EdNet-KT1.[5] These publicly available datasets are acquired from the educational tutoring platforms.

- **ASSIST09**: is an open dataset collected from the Assistments online tutoring systems, which is widely used by researchers for KT and student modeling. Specifically, we choose the corrected version fixing the data modeling issues [40].
- **ASSIST12**: comes from the same platform as ASSIST09, which covers response records of students during the year 2012-2013. This dataset has one concept for one question.
- **ASSISTchall**: is attained from ASSISTment 2017 data mining competition and gathered from a longitudinal study that tracks students from their use of Assistment.
- **Statics2011**: is obtained from a college engineering statics course at Carnegie Mellon University and contains the maximum number of average records per student.
- **EdNet-KT1**: the largest comprehensive interaction dataset, is provided by Santa [41]. We reduce the size of EdNet-KT1 dataset by randomly sampling 5000 users among 600K users.

Note that during the dataset pre-processing, we exclude students from ASSIST09, ASSIST12 and ASSISTchall who have fewer than 15 response logs to ensure that each student has sufficient data for knowledge tracing. Furthermore, we merely choose students with more than 10 response logs for Statics2011 and EdNet-KT1. The detailed statistics of these datasets are shown in Table 2.

5.1.2. Implementation settings

We set the embedding size $d = 128$ [23] and the number of largest singular values $\upsilon = 5$ [37]. Next, the construction threshold η is set to 0.08. The number of layers l is set to 1 for GNN_{int} and 2 for GNN_{str}. The balance coefficient α is set to 0.6. Besides, we initialize the whole parameters with Xavier initialization. Meanwhile, we utilize the Adam algorithm [42] for optimization with an initial learning rate of 0.001 and a batch size of 64. Our method is implemented using Pytorch and we perform all experiments

[1] https://sites.google.com/site/assistmentsdata/home/assistment-2009-2010-data.
[2] https://sites.google.com/site/assistmentsdata/home/2012-13-school-data-with-afect.
[3] https://sites.google.com/view/assistmentsdatamining/dataset.
[4] https://pslcdatashop.web.cmu.edu/DatasetInfo?datasetId=507.
[5] http://ednet-leaderboard.s3-website-ap-northeast-1.amazonaws.com/.

Table 2
Summary statistics of five datasets.

Datasets	ASSIST09	ASSIST12	ASSISTchall	Statics2011	EdNet-KT1
Students	4,163	22,591	1,709	332	5,000
Questions	17,746	52,855	2,210	1,223	12,081
Concepts	123	265	101	156	189
Records	278,868	2,682,439	351,820	189,292	632,230
Avg. Length	66.99	118.73	205.86	570.16	126.45

on a Linux server with Quadro RTX 6000 GPU with 24 GB of video memory. Moreover, to make the results comparable, the other hyperparameters of all the baselines are carefully tuned to achieve their best performance.

5.1.3. Baselines and evaluation metrics

The mainstream KT methods are selected to verify the effectiveness of our proposed method. Specifically, we divide these methods into two categories: response concatenation methods (DKT, DHKT, CoKT) and response initialization methods (DKVMN, SKVMN, SAKT, CL4KT). To effectively show the improvement of QRCL, we extend DKT, CoKT, DKVMN, SAKT, and CL4KT. DHKT and SKVMN are excluded from our analysis is due to their similarity in network structures to DKT and DKVMN, respectively. In addition, to ensure fairness, all methods were tuned to achieve the best performance. The details are described as follows:

- **DKT**[3]: is an implementation using an LSTM network that leverages a series of one-hot representations of questions and trains a unified network for predicting students' next interaction on certain question.
- **DHKT**[11]: incorporates the hierarchical information between questions and concepts into knowledge tracing model and effectively obtains meaningful representations of questions.
- **CoKT**[7]: obtains the inter-student information by retrieving the records and subsequences of the students who have similar question-answering experiences and combines it with intra-student information to make predictions.
- **DKVMN**[4]: utilizes a key-value memory network to store underlying concept representations and states.
- **SKVMN**[13]: improves DKVMN by using a modified LSTM with hops, called Hop-LSTM, to better discover the correlation between latent concepts and questions for prediction.
- **SAKT**[14]: is a recently proposed deep-learning method based on self-attention mechanism to capture the relation between interactions.
- **CL4KT**[8]: devises a contrastive learning method alleviating the issue of sparse interaction between students and questions by means of effective self-supervised signals.

Moreover, we evaluate the methods on these datasets from classification perspective. Hence, we conduct the experiments under Accuracy (ACC, the higher the better) metric and Area Under the ROC Curve (AUC, the higher the better) metric for model evaluation.

5.2. Performance comparison (RQ1)

To answer the RQ1, we report the averages and standard deviations of the performance across conducting five repetitions on the prediction task in Table 3. In particular, the best performance and the second-best performance models are denoted in bold and underlined, respectively. Moreover, based on the final results in Table 3, Fig. 6 vividly shows our improvements on ACC and AUC compared to the original method after using the QRCL. Therefore, we have the following interesting observations.

Firstly, according to the results in Table 3, there are methods that perform favorably performance and approaches that exhibit inferior performance under both categories (i.e., response concatenation and response initialization), indicating that generating question-response representations using these two methods is a viable option. However, the QRCL extended methods consistently perform better than original baselines on all datasets. This means the proposed QRCL method can obtain the superior question-response representations, thus effectively enhancing the performance of downstream KT models. Secondly, from Fig. 6, we can observe that the QRCL extended methods have the best improvement results on ASSISTchall on ACC and AUC, compared to ASSIST09 and EdNet-KT1. One possible reason for this is that the average number of questions that students interact with on Statics2011 is large, which results in a denser co-response relation graph. As a result, the corresponding views can contain more neighbor information, leading to an improvement in the final question-response representations. Lastly, we notice that QRCL+DKT, QRCL+DKVMN and QRCL+SAKT methods have greater improvement than QRCL+CoKT as well as QRCL+CL4KT. We assume that the original CoKT and CL4KT models with multiple complex modules are not sensitive to changes of question-response representations. Up to now, RQ1 has been answered in detail.

5.3. Ablation experiments (RQ2)

To further investigate the effectiveness of each component in QRCL on the performance prediction, we perform ablation studies comparing QRCL with its two variants. Specifically, we leverage QRCL+CoKT and QRCL+CL4KT on five datasets. Here, the reason for choosing these two methods is that they have the best performance in their own categories. The details are illustrated below:

Table 3
Comparison results on student performance prediction on five datasets.

Methods		ASSIST09		ASSIST12		ASSISTchall		Statics2011		EdNet-KT1	
		ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC	ACC	AUC
Response concatenation	DKT	0.7362±0.0034	0.7561±0.0029	0.7286±0.0029	0.7359±0.0033	0.7098±0.0035	0.7178±0.0032	0.7329±0.0028	0.7514±0.0030	0.7141±0.0033	0.7438±0.0034
	DHKT	0.7623±0.0031	0.7786±0.0030	0.7557±0.0025	0.7641±0.0026	0.7123±0.0033	0.7221±0.0031	0.7591±0.0036	0.7764±0.0032	0.7317±0.0031	0.7455±0.0029
	CoKT	0.7724±0.0023	0.7893±0.0021	0.7651±0.0020	0.7761±0.0021	0.7442±0.0022	0.7606±0.0024	0.7698±0.0026	0.7845±0.0025	0.7587±0.0023	0.7692±0.0021
Response initialization	DKVMN	0.7403±0.0042	0.7525±0.0036	0.7301±0.0031	0.7501±0.0034	0.7103±0.0036	0.7187±0.0034	0.7362±0.0038	0.7542±0.0035	0.7234±0.0039	0.7496±0.0041
	SKVMN	0.7501±0.0038	0.7721±0.0037	0.7449±0.0032	0.7597±0.0033	0.7133±0.0035	0.7295±0.0037	0.7486±0.0036	0.7645±0.0042	0.7312±0.0043	0.7518±0.0039
	SAKT	0.7681±0.0026	0.7861±0.0028	0.7569±0.0023	0.7758±0.0020	0.7144±0.0024	0.7261±0.0027	0.7613±0.0028	0.7837±0.0027	0.7486±0.0028	0.7665±0.0025
	CL4KT	0.7765±0.0023	0.7926±0.0025	0.7661±0.0019	0.7784±0.0021	0.7423±0.0023	0.7599±0.0022	0.7702±0.0029	0.7872±0.0024	0.7618±0.0025	0.7705±0.0027
QRCL extended	QRCL+DKT	0.7694±0.0031	0.7944±0.0032	0.7660±0.0024	0.7811±0.0025	0.7571±0.0031	0.7764±0.0030	0.7871±0.0027	0.8187±0.0026	0.7529±0.0028	0.7906±0.0031
	QRCL+CoKT	0.7816±0.0021	0.8051±0.0020	0.7762±0.0019	0.7955±0.0018	0.7693±0.0021	0.7988±0.0022	0.7997±0.0023	0.8278±0.0024	0.7722±0.0020	0.7912±0.0019
	QRCL+DKVMN	0.7704±0.0037	0.7888±0.0036	0.7615±0.0030	0.7899±0.0031	0.7558±0.0033	0.7770±0.0031	0.7880±0.0035	0.8197±0.0032	0.7569±0.0036	0.7899±0.0038
	QRCL+SAKT	**0.7946±0.0023**	**0.8202±0.0024**	**0.7852±0.0020**	**0.8111±0.0019**	0.7564±0.0020	0.7781±0.0025	**0.8091±0.0024**	**0.8429±0.0025**	0.7763±0.0026	**0.8052±0.0023**
	QRCL+CL4KT	0.7878±0.0022	0.8107±0.0023	0.7820±0.0017	0.8016±0.0018	**0.7707±0.0022**	**0.8027±0.0020**	0.8041±0.0026	0.8349±0.0023	**0.7792±0.0023**	0.7958±0.0024

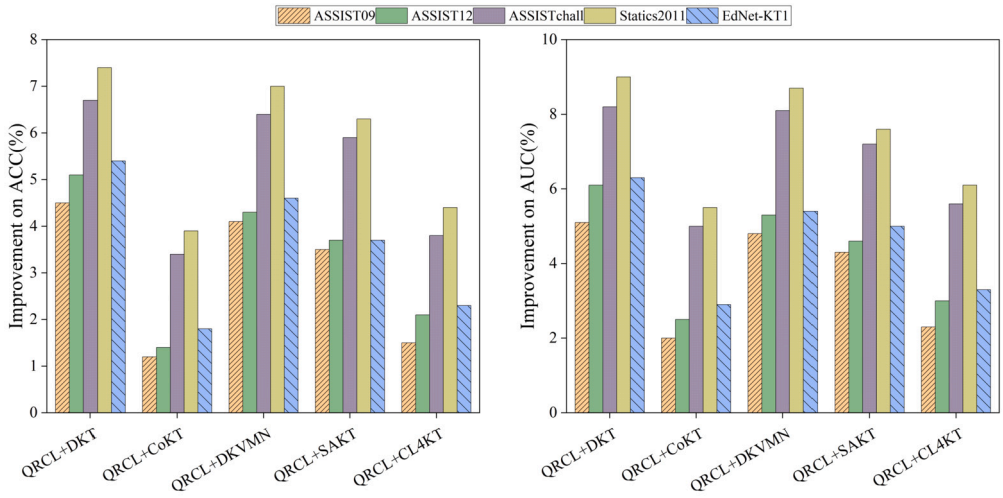


Fig. 6. The improvement of different QRCL extended method on five datasets.

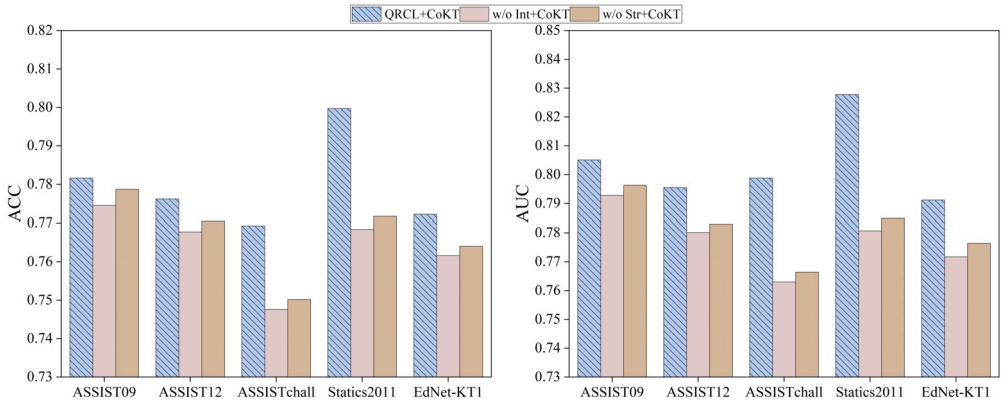


Fig. 7. Performance comparison of ablation study of QRCL+CoKT on five datasets.

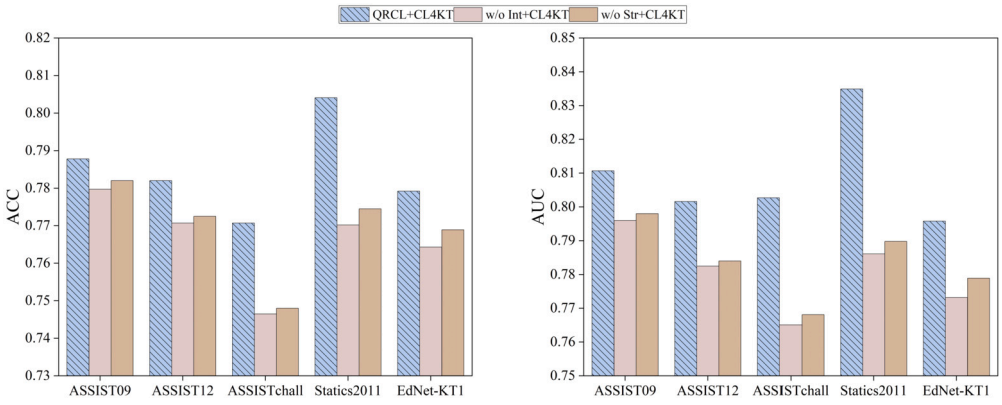


Fig. 8. Performance comparison of ablation study of QRCL+CL4KT on five datasets.

w/o Int, which removes the augmented fold views obtained by means of the SVD method and only performs structure-level contrastive learning to generate the question-response representations.

w/o Str, which removes the co-response relation views acquired from learning records and only performs interaction-level contrastive learning to generate the question-response representations.

To be fair, except for the changes mentioned above, the other parts of aforementioned methods and implementation settings remain identical. We show the results in Fig. 7 and Fig. 8, with some observations. First, removing either augmented fold views

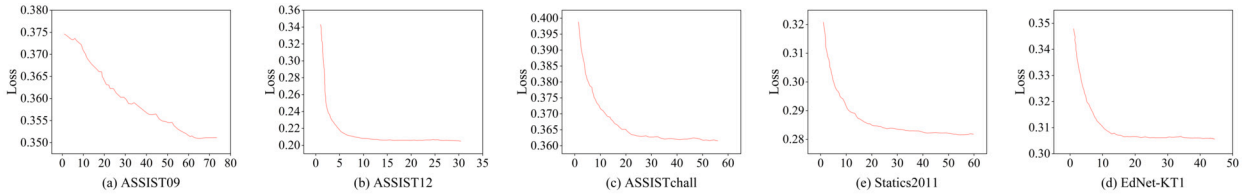


Fig. 9. Convergence curve on five datasets.

Table 4
Comparisons of runtime (in seconds). The results illustrate the time cost associated with question-response representation per epoch.

Datasets	QRCL	DKT	CoKT	DKVMN	SAKT	CL4KT
ASSIST09	52.82	10.51	18.74	34.86	31.12	33.74
ASSIST12	61.13	16.35	23.16	42.47	34.59	46.02
ASSISTchall	32.49	6.37	12.03	22.08	19.32	26.81
Statics2011	26.77	4.08	9.32	18.14	15.33	21.35
EdNet-KT1	49.04	9.56	17.35	31.23	29.87	32.59

or co-response relation views has a harmful impact on predicting the performance of students accurately, which implies that these views obtained with the SVD approach or mining learning records are indispensable for question-response representations. Second, augmented fold views play a crucial role in performance prediction, and their elimination can result in a significant drop in performance. This suggests that interaction-level contrastive learning with augmented fold views contributes more to the question-response representations, ultimately leading to better predictions.

5.4. Model benefit analysis (RQ3)

In this section, we perform convergence and time complexity analysis on five datasets to further verify the efficiency of our method.

5.4.1. Convergence analysis

We investigate the convergence property of QRCL method in Fig. 9. Along the training process, QRCL reaches convergence at a certain number of epochs on the five datasets. In particular, the model achieves the fastest convergence rate (epoch 12) on the ASSIST12 compared with other datasets. The possible reason is that QRCL can learn more critical information and change in three types of views on ASSIST09 containing most records (i.e., 2,682,439), thereby enabling QRCL to converge quickly. This observation suggests that exploring dual-level contrastive learning based on the delicately designed contrastive rule is valuable and useful for question-response representation generation.

5.4.2. Comparison with the efficiency of various baselines

To better understand our method, we compare the runtimes of the different methods in Table 4. It is worth noting that QRCL is designed to improve downstream KT methods, and it fails to directly compare runtimes with these KT methods. Therefore, we only obtain the time cost related to question-response representation in these downstream KT methods for achieving comparison with QRCL. It is clear that QRCL incurs a relatively high time but affordable cost on all five datasets, which is unavoidable due to the structure of our method. However, upon closer examination of the performance prediction results (refer to Table 3), QRCL demonstrates exceptional proficiency in achieving a balance between time consumption and performance enhancement.

5.5. Sensitivity analysis (RQ4)

In this section, to understand the significant hyperparameters of QRCL method in more profound detail, we conduct the sensitivity analysis of three significant hyperparameters using QRCL+DKT method on ASSIST09, including the construction threshold η of co-response relation graph, the number of layers l of GNN_{int} and GNN_{str}, and the balance coefficient α between interaction-level contrastive loss and structure-level contrastive loss. The other hyperparameters remain unchanged when we study the influence of one hyperparameter in the experiments.

5.5.1. Construction threshold of co-response relation graph

The construction threshold η is the key to the co-response relation graph, which controls the number of edges in the graph. To further investigate its effectiveness, we set the other parameters to the best value while adjusting it in the range of $\{0.01, 0.02, 0.04, 0.06, 0.08, 0.1, 0.2, 0.3\}$. The experimental results in Fig. 10 suggest some interesting conclusions.

Firstly, we can easily see that the performance continues to improve as the construction threshold increases at the beginning. One possible reason for this is that a smaller construction threshold leads to a denser co-response relation graph, resulting in

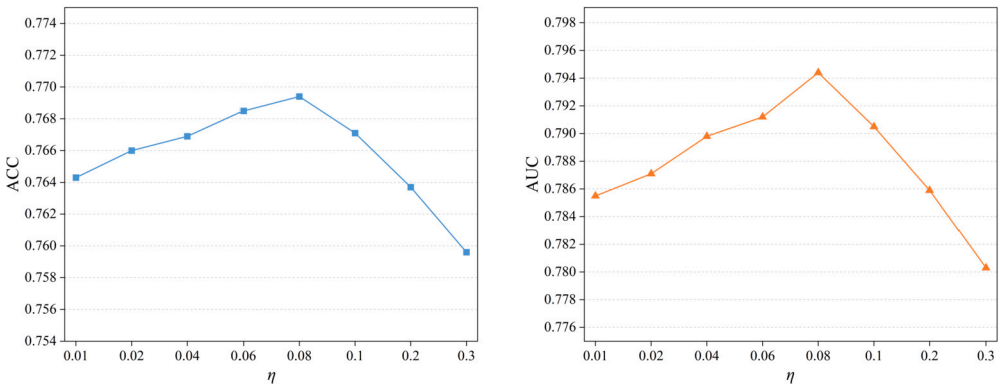


Fig. 10. Prediction performance of construction threshold of QRCL+DKT on ASSIST09.

Table 5
The effect of the numbers of graph convolution layers. LN denotes the layer numbers. $I_i_S_j$ means i graph convolution layers in GNN_{int} and j graph convolution layers in GNN_{str}. The best performance is denoted in bold.

LN	ACC	AUC	*LN*	ACC	AUC	*LN*	ACC	AUC
$I_1_S_1$	0.7636	0.7927	$I_2_S_1$	0.7561	0.7815	$I_3_S_1$	0.7544	0.7788
$I_1_S_2$	**0.7694**	**0.7944**	$I_2_S_2$	0.7513	0.7769	$I_3_S_2$	0.7469	0.7712
$I_1_S_3$	0.7599	0.7863	$I_2_S_3$	0.7451	0.7707	$I_3_S_3$	0.7423	0.7686

less differentiation in the learned question-response representations and ultimately weakening the performance. After that, the QRCL+DKT has the best performance on both ACC and AUC when $\eta = 0.08$. At last, as the construction threshold becomes larger, the performance obviously decreases. And we argue that the learned question-response representations fail to acquire sufficient information about neighbors due to the sparsity of the edges in the co-response relation graph. Therefore, to maintain the optimal performance, we set $\eta = 0.08$ in this paper.

5.5.2. The effect of layer numbers

As the number of graph convolution layers used in GNN_{int} and GNN_{str} are critical hyperparameters, we examine their impact in this subsection. To be specific, we search for the values of the layer numbers in the range of $\{1, 2, 3\}$. The results are summarized in Table 5. We can observe that $I_1_S_2$ setup works best on prediction task, which indicates this structure has the optimal capacity of information aggregation. However, the performance decreases to some extent when the layer numbers in GNN_{int} and GNN_{str} conformably become larger, i.e., $I_3_S_3$ has the worst result. The possible reason is that the multi-layer embedding propagation fully captures the high-order neighbor information, but doing so fails to obtain individual question-response representation of each question, thus impairing the prediction performance.

5.5.3. Balance coefficient of dual-level contrastive loss

The balance coefficient α indicates which kind of contrastive learning the QRCL prefers, which is essential for model learning. Hence, we set α in the range of $\{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1\}$ to explore the optimal performance. Note that a larger balance coefficient reflects that QRCL pays more attention to interaction-level contrastive learning. Fig. 11 shows the corresponding results, and we can see that the balance coefficient that is either too small or large can have a noticeable impact on the performance. Besides, the QRCL+DKT achieves the best performance at $\alpha = 0.6$, which demonstrates that interaction-level contrastive learning is more helpful.

5.6. Case study (RQ5)

In this subsection, our objective is to investigate the characteristics of the learned question-response representations. To accomplish this, we visualize the question-response embeddings of our model and the comparative method, and provide detailed similarity analysis to shed light on their plausibility.

5.6.1. Visualization

To enhance our understanding of the learned embeddings, we randomly choose a set of 3068 questions that encompass six distinct concepts in ASSIST09. Using t-SNE [43], we map the embeddings of these questions into a two-dimensional space, as depicted in Fig. 12. These question-response representations are derived from both the DKT and QRCL+DKT models, as explained in [23]. Besides, we specifically demonstrate two question-response representations from QRCL, representing correct and incorrect responses, to offer clear visualizations of the outcomes. We have the following observations in Fig. 12.

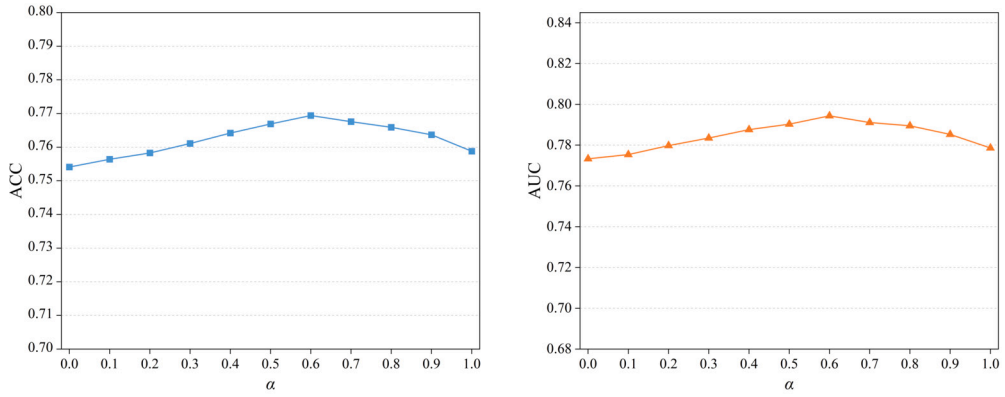


Fig. 11. Prediction performance of balance coefficient of QRCL+DKT on ASSIST09.

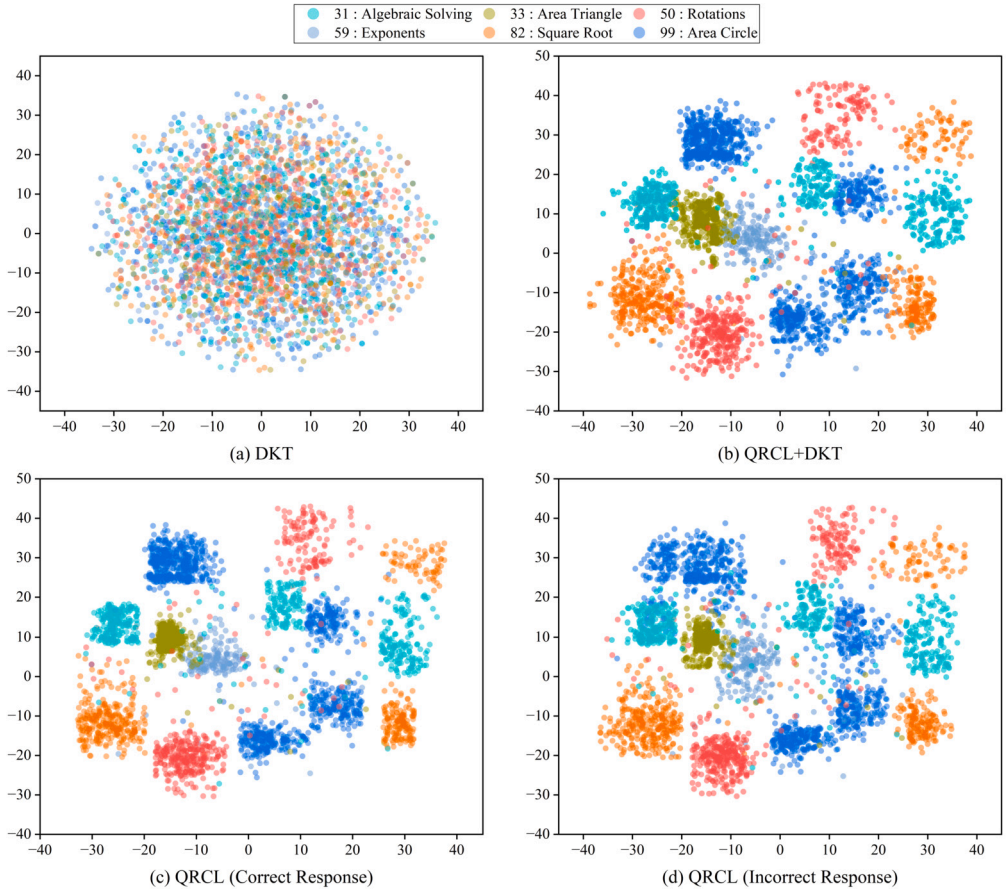


Fig. 12. The visualization of question-response representations of chosen questions on ASSIST09. Questions with the same concept are marked in same color.

As depicted in Fig. 12(a), the representations in DKT are scattered randomly, indicating a failure to capture the relationships between questions and concepts effectively. However, Fig. 12(b) demonstrates that QRCL+DKT successfully learns question-response representations, where questions belonging to the same concept are grouped together in distinct clusters. Furthermore, we observe that the question-response representations obtained by QRCL for correct and incorrect responses, as shown in Fig. 12(c) and Fig. 12(d), follow a similar trend to QRCL+DKT, albeit with some differences between them. This suggests that QRCL can learn more meaningful question-response representations that are specifically related to correct and incorrect responses.

5.6.2. Similarity

To conduct a detailed analysis of the similarity between question-response representations in QRCL under correct and incorrect responses, we select a subset of 10 questions that encompass the previously mentioned six concepts. Table 6 outlines the inclusion

Table 6
The inclusion relations between concepts and the chosen questions.

Concepts	Related Questions (Order)
31: Algebraic Solving	q_{5425} (1), q_{5600} (2), q_{5791} (3)
33: Area Triangle	q_{6161} (4)
50: Rotations	q_{268} (5)
59: Exponents	q_{8368} (6), q_{10900} (7)
82: Square Root	q_{12385} (8), q_{12870} (9)
99: Area Circle	q_{15216} (10)

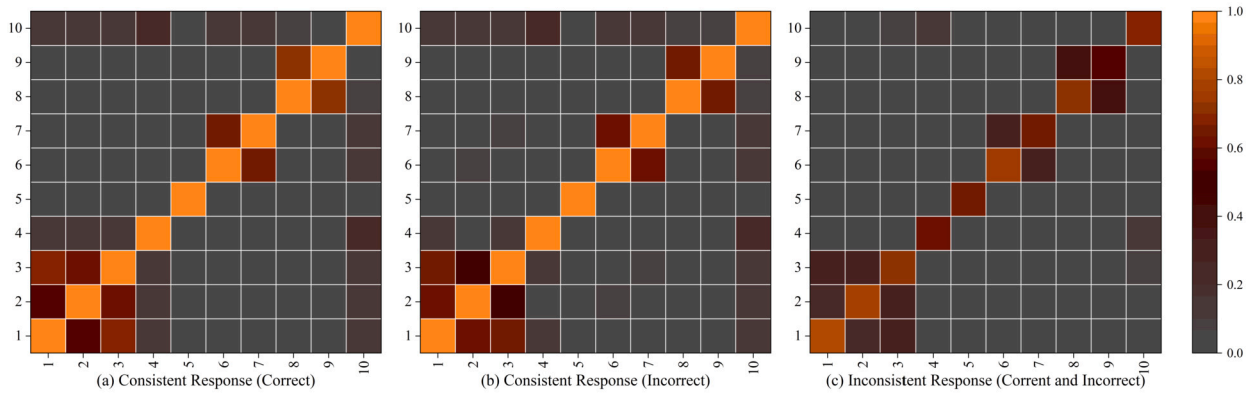


Fig. 13. Visualization of similarity between question-response representations under consistent and inconsistent responses.

relations between these concepts and the corresponding questions. Using cosine similarity as the metric, we calculate the pairwise similarities between the representations, as illustrated in Fig. 13. By examining the results, we make the following intriguing observations.

Firstly, as depicted in Fig. 13, we observe that questions sharing the same concept exhibit considerable similarity, such as the similarities between any two of the questions q_{5425}, q_{5600} and q_{5791}. However, it is worth noting that the similarity between these questions varies, which can be attributed to differences in their difficulty levels. Moreover, questions with similar concepts demonstrate higher similarity compared to questions with unrelated concepts. For example, the similarity between q_{6161} and q_{15216} is higher than the similarity between q_{6161} and q_{8368} or q_{6161} and q_{10900}, aligning with our expectations. Secondly, we observe that the distribution of similarity between questions in Fig. 13(a) and Fig. 13(b) is similar, indicating the reasonableness of the question-response representations under the two responses. Lastly, we notice that the similarity between questions in Fig. 13(c) is lower than in Fig. 13(a) and Fig. 13(b), particularly evident in the secondary diagonal representing the similarity between the same questions. This indicates that the similarity between questions under inconsistent responses is lower than that under consistent responses, thus validating that QRCL acquires two distinct question-response representations.

6. Conclusions

In this paper, we present a novel QRCL method that generates question-response representations in a self-supervised manner for knowledge tracing. To be specific, our approach achieves better question-response representation learning from two dimensions: (1) QRCL considers three types of views, containing the native fold views derived from the native interaction graphs, the augmented fold views attained by contrastive augmentation with SVD, and the co-response relation views acquired with the help of connections among questions. (2) QRCL performs dual-level contrastive learning among the above three kinds of views according to delicately designed contrastive rules, learning the final question-response representations, and further improving the prediction performance of KT methods. Extensive experiments conducted on five real-world datasets demonstrate QRCL can effectively enable KT methods to predict students' performance. In future work, we would like to explore the specific relations between the KT methods with different structures and question-response representation learning. Meanwhile, we intend to leverage the diffusion model to effectively obtain better question-response representations.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All datasets in this paper are publicly available.

Acknowledgements

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